

Wednesday Warm Up: Unit 0, Review of 135A

Prof. Jordan C. Hanson

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1 Memory Bank

- $\vec{v}_{\text{ave}} = \Delta \vec{x} / \Delta t$... Definition of average velocity involving vectors.
- $x(t) = \frac{1}{2}at^2 + v_i t + x_i$... One-dimensional displacement with constant acceleration.
- $v(t) = v_i + at$... One-dimensional velocity with constant acceleration.
- $\vec{F}_{\text{net}} = m\vec{a}$... Newton's 2nd Law.
- $W = \vec{F} \cdot \vec{x}$... Definition of work involving vectors.
- $W = Fx \cos \theta$... Definition of work, with θ as angle between force and displacement.
- $KE = \frac{1}{2}mv^2$... Kinetic energy.
- $W = \Delta KE = KE_f - KE_i$... Work energy theorem.
- $KE_i + PE_i = KE_f + PE_f$... Energy conservation.
- $\vec{p} = m\vec{v}$... Definition of momentum involving vectors.
- $\vec{p}_{\text{tot},i} = \vec{p}_{\text{tot},f}$... Conservation of momentum.
- $\tau = I\alpha$... Newton's 2nd Law for rotating objects, with torque, moment of inertia, and angular acceleration.
- $KE_{\text{rot}} = \frac{1}{2}I\omega^2$... Rotational kinetic energy.
- $L = I\omega$... Angular momentum.

2 Warm-Up Exercises

1. Suppose a ship sails at 20 km per hour for 3 hours to the West, and then at 20 km per hour for 2 hours to the South. (a) Assuming that the ship starts at the origin of a 2D coordinate system, what is the final location of the ship? (b) What is the average velocity?
2. Suppose an athlete starts a race from rest at $t = 0$ with a constant acceleration of $a = 2.5 \text{ m s}^{-2}$. (a) How long before the speed of the runner is 5 m s^{-1} ? (b) If the runner has a mass of 60 kg, what is the kinetic energy? (c) How much work or energy was required to reach this velocity?
3. Suppose two children each pull a toy in opposite directions. One pulls to the right with a force of 10 N, while the other pulls to the left with a force of 8 N. The toy weighs 0.5 kg. (a) What is the magnitude and direction of the acceleration of the toy? (b) If the toy begins at rest, where is the toy after 2 seconds? (c) If the kids drop this same toy from a height of 10 meters, what will be the final velocity just before it hits the ground?
4. Suppose a physical therapy patient is asked to shove a medicine ball forward off the edge of a table to help rebuild the strength of their shoulders. The medicine ball weighs 7 kg. (a) If the patient is able to give the ball a speed of 1 m s^{-1} , what is the momentum of the ball? (b) If the patient gives the ball the same momentum by rolling it, and it strikes elastically a ball with a mass of 3.5 kg, what will be the velocity of the second ball?
5. Suppose the medicine ball in the previous problem has a mass of 3.5 kg, and a diameter of 10 cm. The moment of inertia for a solid sphere is $I = \frac{2}{5}mr^2$. (a) What is the angular momentum of the ball if it is spun at 1 rotation per second? (b) What is the rotational kinetic energy?